Data Warehouse & Business Intelligence

Topic: Gender Analysis in European Higher Education (2013–2023)

Course: Data Warehouse and Business Intelligence
Academic year: 2025–2026
Student: Pedro Ferreira

Conteúdo

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Part 1 — Theoretical Component

1. Introduction

EuroEdu Analytics Center is a fictional entity created within the scope of this project, representing a European center specialized in the collection, management, and analysis of data on education and social equality. It is an independent, non-profit organization focused on producing statistical analyses that support informed decision-making in the educational sector.

The center is composed of a multidisciplinary team of data analysts, statisticians, and information systems specialists, and benefits from access to public databases and academic collaborations that strengthen the reliability of its analyses. European higher education is one of its priority areas of study, given the increasing importance of understanding how men's and women's participation is distributed across countries, study cycles, and fields of education.

Despite the breadth and quality of the PORDATA database, its dispersed organization across multiple categories makes immediate comparative analysis difficult. In this context, implementing a Data Warehouse (DW) solution combined with a Business Intelligence (BI) system allows the data to be consolidated, normalized, and explored systematically and interactively.

This report describes all phases of the project—from conceptual definition to practical construction of the model—focused on analysing gender parity in European higher education between 2013 and 2023.

2. Requirements Analysis

Requirements analysis is the essential starting point to ensure that a BI solution effectively meets the organization's needs. In the case of EuroEdu Analytics Center, the objective is to obtain a clear, structured, and comparable view of gender distribution in European higher education, enabling the identification of patterns, asymmetries, and time trends based on concrete data.

2.1 Business Objective

The main objective is to build an analytical solution capable of:

- Assessing the level of gender parity in European higher education.
- Comparing realities between countries, fields of education, and study cycles.
- Identifying trends in evolution between 2013 and 2023.
- Supporting the design of more informed and equitable educational policies.

The solution must be flexible, adapting to the needs of different user profiles, both for high-level analysis and more detailed queries.

3. Key Performance Indicators (KPIs)

KPIs were defined and grouped into four main categories to answer the core analysis questions.

3.1 Global KPIs — Overall Parity Status

These indicators provide an overall view of gender distribution:

- Total Students
- Total Men
- Total Women
- Percentage of Men
- Percentage of Women
- Percentage Difference Between Genders

They allow quick assessment of whether there is an approximate 50/50 balance or relevant deviations.

3.2 Comparative KPIs — Where Are the Deviations?

These indicators detail distribution across specific categories:

- Percentage of Men/Women by Country
- Percentage by Study Cycle
- Percentage by Field of Education

This analysis helps distinguish whether inequalities are:

- Cultural (country-associated)
- Structural (related to level of education)
- Vocational (linked to fields of study)

3.3 Temporal KPIs — How Did Parity Evolve?

Indicators that analyse change over the decade:

- Annual variation in the percentage of women
- Annual variation in the percentage of men

They allow observation of trends such as growth, stability, or setbacks.

3.4 Contribution KPIs — Who Weighs Most in the European Total?

Indicators that evaluate each country's weight in the European panorama:

- Contribution of Men by Country
- Contribution of Women by Country

They are useful to identify countries with greater statistical influence due to population size.

4. Stakeholders

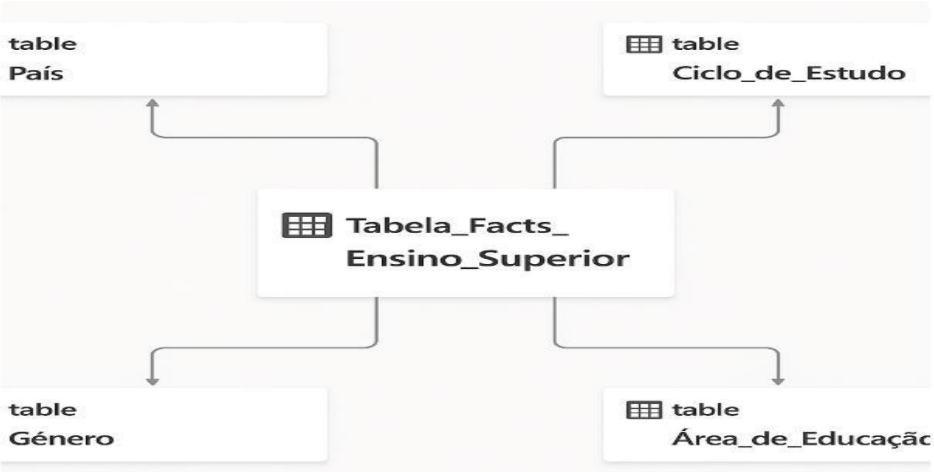
The system was designed to meet the needs of several user profiles:

- Management: requires global indicators and strategic comparisons.
- Statistical Research Team: needs detailed and segmented analyses.
- BI Team: responsible for ETL processes, validation, and system performance.

- End Users/Analysts: use interactive dashboards to generate reports and recommendations.

Clearly identifying these profiles helped align the level of detail of KPIs with each group's operational reality.

5. Conceptual and Logical Modelling



5.1 Star Schema

The adopted dimensional model follows a star schema, as it is well-suited to simple BI systems, efficient, and easy to interpret—especially in tools like Power BI. Although in practice a single consolidated table was used (since PORDATA data is already denormalised), the conceptual model maintains the formal DW structure: one central fact table and several contextual dimensions.

5.2 Fact Table

The `FactEnsinoSuperior` table includes the following columns:

- Year
- Country
- Gender
- StudyCycle
- FieldOfEducation
- NumberOfStudents

Each record represents a unique combination of these dimensions, with granularity defined as: Year + Country + Gender + Study Cycle + Field of Education.

5.3 Conceptual Dimensions

The model defines five conceptual dimensions (not materialised as separate tables):

- Country (list of European countries considered)
- Year (2013 to 2023)

- Gender (Male, Female)
- Study Cycle (Short Cycle, Bachelor, Master, Doctorate)
- Field of Education (PORDATA categories)

These dimensions form the theoretical foundation of the model and align with BI modelling best practices.

6. DAX Measures

DAX measures were used instead of calculated columns for three main reasons:

- Measures dynamically respond to applied filters.
- They avoid increasing model size.
- They keep the report lighter and better organised.

Key measures include: Total Students, Total Men, Total Women, Men %, Women %, Percentage Difference Between Genders, and GPI (Gender Parity Index).

7. ETL Processes

The ETL (Extract, Transform, Load) process was carried out in two complementary stages in Excel and Power BI (Power Query), ensuring adequate data preparation for the dimensional model.

7.1 Extraction

Data was obtained from PORDATA in CSV/Excel format, containing aggregated records by year, country, gender, study cycle, field of education, and number of enrolled students.

7.2 Transformation

Stage1—Excel

- Convert CSV to a Table (Get Data).
- Normalise headers (e.g., “Study cycle” → `StudyCycle`).
- Manually filter blank values.
- Validate data types.
- Remove irrelevant or empty columns.

Stage 2 — Power Query (Power BI)

- Confirm data types.
- Create a DAX measure for Total Students excluding blanks (because blanks were not filtered out initially).
- Standardise text for consistent filtering.
- Remove duplicates.

Key decision: all calculations were implemented via DAX measures, avoiding derived columns.

7.3 Loading

The final cleaned and structured table was loaded into Power BI. The next steps included creating measures, validating filters and visual interactions, and preparing dashboards.

8. Data Governance and Security

Even though the data is public and aggregated, the implementation followed essential data governance principles.

- Access Control (RBAC): In Power BI Service, roles could be defined such as Administrator (model management and updates), Editor (report creation), and Viewer (dashboard consumption).
- Quality and Traceability: Document Power Query steps, consistent naming for columns and measures, cross-validation with original PORDATA data, and annual verification/updates.

9. Model Evolution

The model is prepared for:

- Adding new years.
- Creating new DAX measures.
- Integrating new fields of study.

10. Theoretical Conclusion

The developed solution transformed dispersed PORDATA data into an organised structure, validating in practice the concepts covered in the DW and BI course. The conceptual star schema proved effective and aligned with analytical objectives, with sufficient granularity for detailed comparisons. Using DAX measures exclusively kept the model light and responsive.

From a technical perspective, the ETL process reinforced the importance of cleaning, normalising, and validating data before loading. Even with a well-structured public source, careful preparation was essential to ensure analytical quality. Through the created dashboards, it was possible to analyse gender differences over time and between countries, study cycles, and fields of education. The KPIs enabled clear and evidence-based interpretations, providing a solid foundation for reports and recommendations.

Part 2 — Practical Component

1. Topic Selection and Data Source

The practical component of this project consisted of implementing the previously described DW/BI solution, applied to the topic **Gender Analysis in European Higher Education (2013–2023)**. This analysis was conducted in the context of the fictional organisation **EuroEdu Analytics Center**, which uses this system to monitor gender parity in higher education and generate comparative studies across countries, study cycles, and fields of education.

The selected data source was **PORDATA**, provided in CSV/Excel format. The dataset aggregates information by **Year**, **Country**, **Gender**, **Study Cycle**, **Field of Education**, and **Number of Enrolled Students**. Its use ensures reliability, consistency, and comparability across European countries, meeting the quality criteria required for a BI project based on public data.

2. Data Preparation and Cleaning (ETL)

The ETL process was carried out in two main phases: first in **Excel**, then in **Power BI Power Query**. The objective was to ensure that the data was cleaned, structured, and ready for dimensional modelling.

2.1 Normalising Column Names and Values

After importing the CSV file into Excel, the data was converted into a structured table, facilitating initial transformations. Actions performed included:

- Header normalisation (e.g., "Study cycle" → `StudyCycle`).
- Text standardisation (case consistency and removal of extra spaces).
- Data type validation and correction: Year and Number of Students as integers; Country, Gender, Study Cycle and Field as text.

This ensured correct interpretation in Power BI and proper functioning of filters, slicers and measures.

2.2 Handling Missing Values

The `Number of Students` column contained some blank/null values for certain combinations. Instead of deleting rows, missing values were handled through **DAX measures**, creating a Total Students measure that sums only valid records and ignores missing values.

Benefits:

- Avoided removing potentially useful data.
- Ensured visuals and KPIs display only valid values.
- Preserved simplicity and model performance.

2.3 Derived Columns vs. Measures

Although derived columns were initially expected, they were not implemented because all analytical needs were aggregated (percentages, ratios, differences). All logic was implemented with **DAX measures**, providing flexibility, reducing model size, and improving organisation.

Total Alunos = [Total Mulheres] + [Total Homens]

Total de Homens=CALCULATE(SUM([Número_de_alunos]),'Tabela_Facts_Ensino_Superior'[género]= "Homens")

Total Mulheres = CALCULATE (SUM ([Número_de_alunos]),'Tabela_Facts_Ensino_Superior'[género] = "Mulheres")

IPG por países: >1 mais mulheres <1 mais Homens = DIVIDE ([Total Mulheres], [Total Homens])

% Homens = DIVIDE ([Total Homens], [Total Alunos]**% Mulheres** = DIVIDE ([Total Mulheres], [Total Alunos])

Diferença entre total de géneros = ABS ([Total Mulheres] - [Total Homens])

Diferença percentual Géneros = ABS ([% Mulheres] - [% Homens])

IPG por países: >1 mais mulheres <1 mais Homens = DIVIDE ([Total Mulheres], [Total Homens])

3. Dimensional Modelling

3.1 Conceptual Model

The model follows star schema principles. However, because PORDATA data is denormalised, the practical implementation used a single consolidated fact table.

The conceptual model includes one fact table (`FactEnsinoSuperior`) and five theoretical dimensions: Country, Year, Gender, Study Cycle and Field of Education.

3.2 Practical Implementation

In practice, the data was loaded into a single table named `FactEnsinoSuperior` with:

- Year
- Country (Europe)
- Gender
- StudyCycle
- FieldOfEducation
- NumberOfStudents

Each line corresponds to a unique combination of these fields, respecting the defined granularity.

3.3 Relationships and Integrity

Because a single consolidated table was used, no PK/FK relationships were required. Validations confirmed: no duplicates at the intended grain, consistent text categories, and integrity ensured by ETL.

4. Building the Data Model in Power BI

4.1 Loading the Prepared Table

The final table was loaded into Power BI as `TabelaFactsEnsinoSuperior`. Data types were confirmed: Year and Number of Students as numeric; Country, Gender, StudyCycle and FieldOfEducation as text.

4.2 Model Integrity Validation

Validations ensured reliability: no unexpected nulls in key fields, no duplicates at the defined granularity, and consistent results under filtering.

4.3 DAX Measures and Calculations

DAX measures were created to operationalise KPIs: totals, percentages, gender differences and GPI. These measures power KPI cards, charts and temporal analyses, supporting dynamic exploration under slicers and filters.

5. Dashboard Development

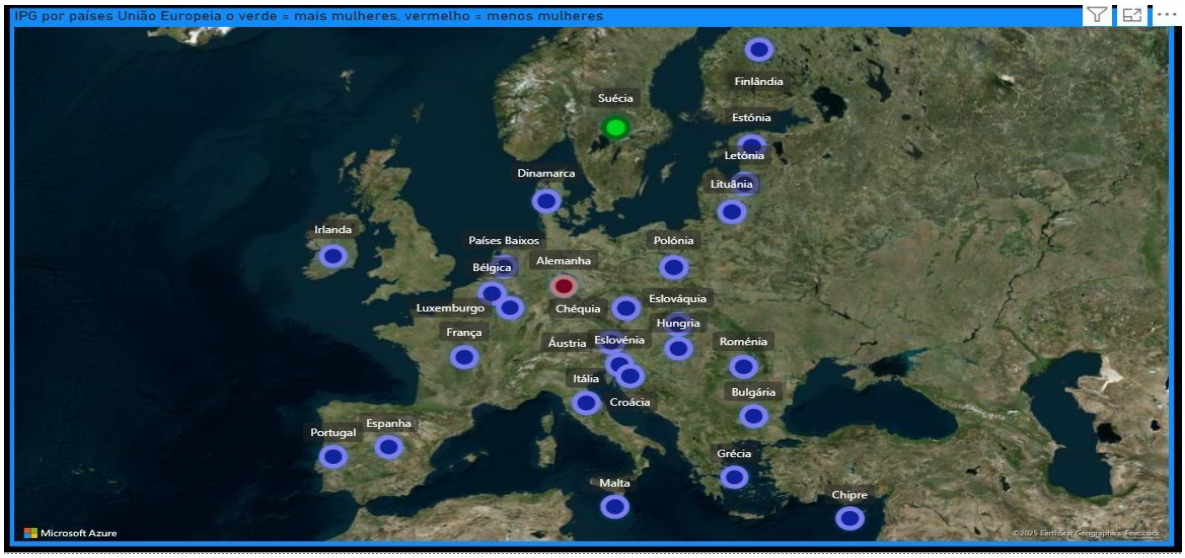
5.1 Overview — Main KPIs

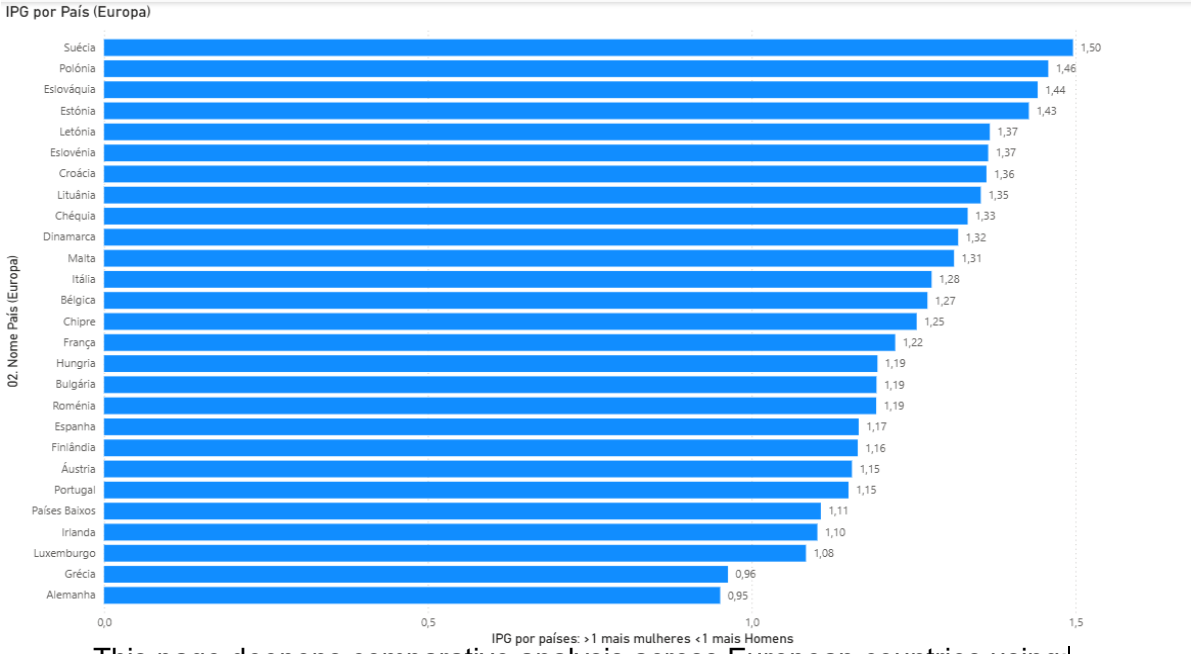
The first page presents the main KPIs and supports a global interpretation:

- Higher number of women enrolled in European higher education.
- Stable proportions: about 54% women and 46% men.
- Approximately 179.39 million men and 211.06 million women between 2013 and 2023, totalling about 390.46 million students.
- Percentage difference between genders around 0.08.

This page serves as the entry point for the analysis, allowing a quick understanding of the overall panorama.

5.2 Comparative Analysis by Country and Category



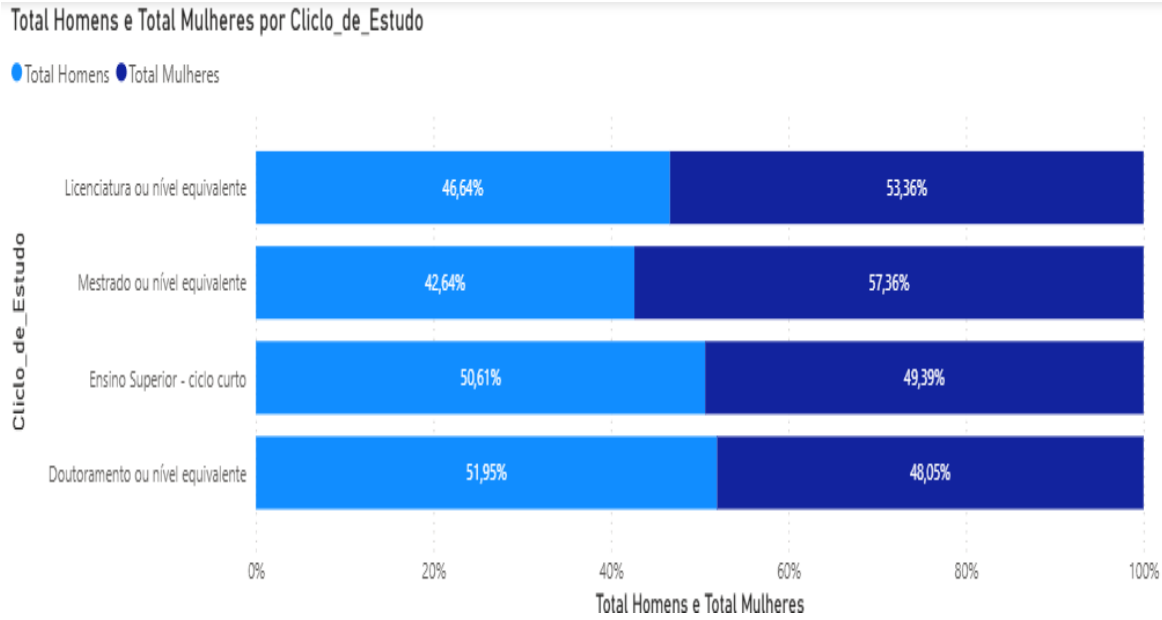


This page deepens comparative analysis across European countries using:

- An interactive map showing GPI distribution across Europe (green = more women; red = more men; blue = intermediate values).
- A horizontal ranking of countries by GPI.

Key readings: Sweden shows the highest female predominance (GPI ≈ 1.50). Germany (≈ 0.95) and Greece (≈ 0.96) are the only countries slightly below parity.

5.3 Study Cycle Analysis

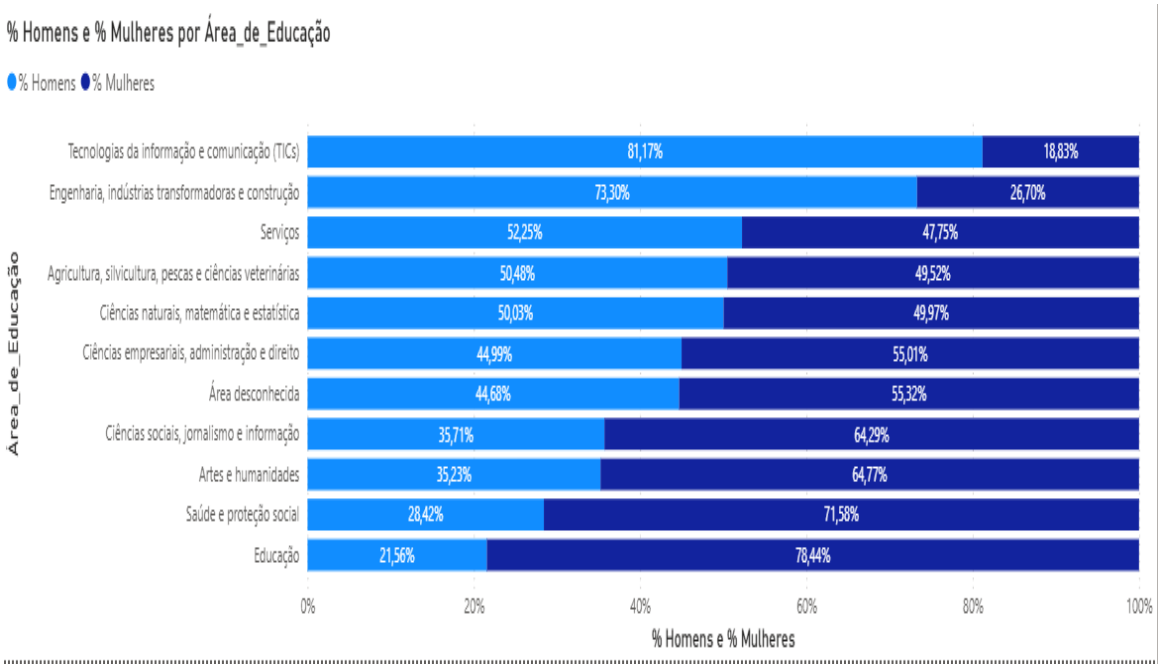


A stacked bar chart shows the distribution of men and women by study cycle:

- Bachelor and Master: higher female shares.
- Short cycle: almost perfect balance.
- Doctorate: slight male predominance.

This highlights moderate structural differences in academic progression.

5.4 Field of Education Analysis



A horizontal chart shows male and female shares across fields.

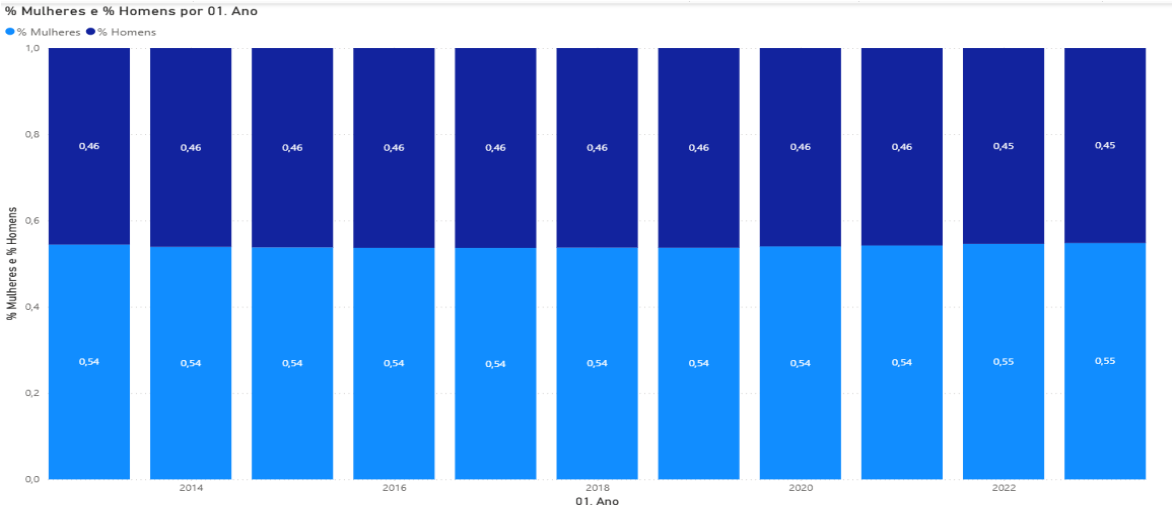
- More women: Education (~78.44%), Health (~71.58%), Arts & Humanities (~64.77%), Social Sciences (~64.29%).
- More men: ICT (~81.17%), Engineering (~73.30%), Natural Sciences (~50.03%).

This view reveals the strongest vocational disparities.

5.5 Temporal Trends

A stacked bar chart of men and women by year shows stable proportions over time: women consistently ~54–55% and men ~45–46%. Minor annual variations exist but there are no major changes.

5.6 Slicers, Filters and Interactivity



All visuals benefit from Power BI interactivity: slicers for Year, Country, Gender, Study Cycle and Field; filters that automatically update KPIs and charts; drill-down and drill-through; and cross-highlighting when selecting elements in maps and rankings.

5.7 Design and Visual Presentation

The design followed a simple, professional line: coherent colour palette (blue, green, grey), clear titles, concise legends, and balanced layout to avoid visual overload.

6. Interpretation and Conclusions

6.1 Main Results

Global analysis indicates a relatively balanced gender distribution with slight female predominance (~54% women, ~46% men). Differences between countries are small and stable, suggesting no severe structural issues—rather patterns linked to national contexts and study choices.

The most visible differences occur by field of education: higher female presence in Education, Health, Arts and Social Sciences; higher male presence in ICT and

Engineering. By study cycle, differences are moderate: higher female presence in Bachelor and Master, and slight male predominance in Doctorates.

6.2 Business Conclusions

Countries with the highest impact on European totals (France, Germany, Italy, Spain) are key for interpreting trends but do not show severe imbalances. Small changes in these countries' GPI values can influence the European average, but within a stable and moderate range.

Over time, results show stability in gender proportions, persistence of vocational patterns by field, absence of abrupt structural changes, and only mild increases/decreases.

7. Strategic Recommendations (with Implementation)

7.1 Encourage diversity in ICT and Engineering (male-dominated fields)

How to implement at institutional or governmental level:

- Vocational guidance programs in secondary schools with workshops led by female students in these areas.
- Mixed technology clubs with hands-on activities (robotics, programming, cybersecurity).
- Thematic scholarships open to all, with targeted outreach to underrepresented groups.
- Partnerships between universities and tech companies for visits, internships and accessible bootcamps.

7.2 Increase male participation in Education and Health (female-dominated fields)

How to implement:

- Communication campaigns highlighting male role models and successful careers in these areas, avoiding stereotypes.
- Mentoring programs with male professionals.
- School sessions on diversity in social and educational professions.
- Short internships/experiences in hospitals, schools and NGOs for secondary students.

7.3 Continuous annual monitoring

How to implement:

- Automatically update the dashboard with annual PORDATA data.
- Create simple alerts in Power BI (e.g., >1% variation in gender share by country or field).
- Produce consistent, short annual reports for management (e.g., two pages).

- Track countries with slight deviations (e.g., Germany, Greece) due to their statistical influence.

8. Final Summary

The analysis showed that European higher education has overall gender balance with small and stable differences, but clear vocational patterns across fields of education. The implemented DW/BI model proved appropriate to consolidate the data, identify trends and support moderate and evidence-based decisions aligned with the educational context.

Sources:

- Support used to build the report structure: ChatGPT and Gemini.
- Data source: PORDATA.
- Conceptual model image: ChatGPT.
- Data model construction table in Power BI: Power BI.
- Dashboard development charts: Power BI.